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# Implementing Lattice-based Cryptosystems for Post-Quantum Cryptography: Addressing Quantum Computing Threats

Mukil LD<sup>a</sup>, Mahi Kolli<sup>a</sup>, Shinu M Rajagopal<sup>a,\*</sup>, Prashanth B N<sup>b</sup>

<sup>a</sup>Department of Computer Science and Engineering, Amrita School of Computing, Bengaluru, Amrita Vishwa Vidyapeetham, India

<sup>b</sup>Department of Mechanical Engineering, Amrita School of Engineering, Bengaluru, Amrita Vishwa Vidyapeetham, India

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## Abstract

Lattice-based cryptography seems to be a good area for the researchers in cryptology because of its vulnerability against quantum computing attacks. Such resources as extensive documentation, theoretical foundations, and practical implementations are provided with respect to some key algorithms, which are Learning With Errors (LWE) and Ring-LWE, and sophisticated ones such as homomorphic encryption. This paper aims to make academic research more practical by providing tools for cryptographers and programmers to experiment with lattice-based cryptography in secure communications and data protection. Thus, we examine and implement Lizard, Ring-Lizard, Kyber, Frodo, and NewHope in JavaScript, which are current postquantum cryptography possibilities due to their applicability and efficiency. In this case quantum computational capabilities are referred to as those that may compromise security thus undermining post-quantum cryptography.

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**Keywords:** Lattice-based cryptography; Post-quantum cryptography; Learning With Errors (LWE); Ring Learning With Errors (Ring-LWE); Frodo; Kyber; Lizard; Ring-Lizard; NewHope; JavaScript implementation.

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## 1. Introduction

Quantum computing has made conventional cryptography methods more vulnerable, necessitating post-quantum cryptographic method construction. Given that quantum computers can tackle the hardest complicated problems exponentially quicker than classical computers, RSA and ECC may soon be outdated. Cryptographic algorithms that can withstand quantum attacks are in high demand due to the imminent threat. Among the different proposed solutions, lattice cryptography has emerged as one of the most promising candidates due to the strong foundation in security and practical efficiency. Lattice-based cryptography draws its security from problems in lattices, which, even for quantum computers, are hard to solve. The main schemes in this domain are the Learning With Errors (LWE) and its variant, Ring-Learning With Errors (Ring-LWE). Both these schemes are built upon the difficulty of solving certain problems in high-dimensional lattices, which provide a high level of security that is conjectured to be secure against quan-

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Corresponding author: [mr\\_shinu@blr.amrita.edu](mailto:mr_shinu@blr.amrita.edu)

tum attacks. LWE and Ring-LWE are both theoretically and practically important, serving as the basis for various cryptographic applications [1-4].

This includes extensive documentation and practical implementations of the foundational schemes for lattice-based cryptography. Learning With Errors (LWE) is a basic lattice-based problem applied to securely construct cryptographic protocols. It is known for its strong security guarantees relying on the difficulty of certain lattice problems that are conjectured to be hard for both classical and quantum computers. Erroneous Ring-Learning Ring-LWE is an optimized variation of LWE that uses rings' algebraic structure to compute more efficiently, making it ideal for practical applications. In addition to these basic techniques, the study provides advanced cryptographic primitives needed for post-quantum computing data and communication security. To secure cloud data processing, homomorphic encryption allows the construction of ciphertexts by their functionality without decrypting the data. Privacy-preserving programs must process sensitive data without exposing it to dangers. Digital signatures, another primitive, verify data integrity and authenticity, two essentials for secure communications. Lattice-based digital signatures, which are secure and efficient, have garnered interest as a good alternative to conventional digital signature methods, which are subject to quantum attacks. [5-9].

Various advanced topics in lattice-based cryptography are also addressed in the paper. This includes practical implementations of fully homomorphic encryption, which is capable of performing arbitrary computations on encrypted data, thus opening up completely new possibilities for secure data processing and analysis. This also includes lattice-based constructions for secure multi-party computation, allowing several parties to compute a joint function of their respective inputs while keeping these inputs private. This is particularly relevant for collaborative scenarios where data privacy is a major concern [16-21] [27].

In addition, this paper explores zero-knowledge proofs—a cryptographic technique that allows one party to demonstrate to another that it indeed knows a value without disclosing the value itself. This is crucial for privacy-preserving protocols and finds applications in secure voting, authentication, and beyond. One of the clear strengths of this paper is the inclusion of advanced cryptographic constructs that make it extremely valuable for researchers and practitioners in the field. On the one hand, this document bridges the divide between theoretical work and practice with tools that can help in the deeper understanding of lattice-based cryptography. The documentation is fine-grained, and the practical implementation in code allows researchers and practitioners to study, implement, and test these cryptographic schemes in real-life situations. Only a practical approach deepens understanding of lattice-based cryptography and allows its use in secure communication, data encryption, zero-knowledge proofs, and secure multi-party computation. We chose an encryption scheme “Lizard” based on the learning with errors and learning with rounding (LWR) problems and its ring variant “ring-Lizard”, a modulo-LWE-based encryption scheme “Kyber”, and two quantum secure key exchange protocols “Frodo” and “NewHope” based on their respective problems. Every cryptosystem above was written in JavaScript.

## 2. Related Works

Several works address post-quantum cryptography challenges, including JavaScript implementations for web and IoT contexts. Significant advances in lattice-based blind signatures, group signatures, key encapsulation mechanisms (KEMs), zero-knowledge proofs, and digital signatures have been demonstrated by works leading to robust and efficient constructions resilient to quantum attacks [1-9].

Yuan et al.[1] addressed the challenges of implementing postquantum encryption schemes and key exchange protocols in JavaScript-enabled platforms. The work is not only concerned with the theoretical aspects of lattice cryptography but also gets into the practical considerations of its real-world deployment. Targeting JavaScript compatibility, the research enables the wide spreading of lattice-based cryptographic algorithms over web applications, IoT devices, and other resource-constrained environments. Hauck et al. [2] review recent works on lattice-based blind signatures, with an emphasis on the improvement of security and efficiency. One revisits existing models, with enhancements using key cryptographic frameworks such as LWE and SIS. Several constructions are explored: lattice-based signatures and their blind versions, improvements in the state of the art of zero-knowledge proofs and homomorphic encryption, security proofs, and concrete instantiations for these models. R. Del Pino et al.[3] present the first lattice-based group signatures, based on zero-knowledge proofs of automorphism stability. The paper uses lattice-based cryptographic constructions with the Short Integer Solution (SIS) and Learning With Errors problems, group signature schemes for

anonymity and traceability, and zero-knowledge proof systems for automorphism stability. Duman et al.[4] bring improvements over the basic lattice-based key encapsulation mechanism using a generic Fujisaki-Okamoto transform with prefix hashing. Lattice-based cryptographic constructs and hashing algorithms are used to propose efficient and safe KEM models. The study compares these models' cryptographic performance.

Lyubashevsky et al.[5] show how advanced cryptographic models can improve the security and efficiency of lattice-based ZKPs for integer relations. The authors create feasible and secure zero-knowledge proofs using lattice-based cryptographic primitives. Lattice-based encryption relies on models like the Short Integer Solution and Learning With Errors. Baum et al.[6] employ homomorphic encryption to secure computation over encrypted data in lattice-based cryptography. PCPs and IOPs are new zero-knowledge proving methods using brief reasoning. They also examine the cryptographic framework's arithmetic circuits, identifying issues and providing solutions to improve protocol performance and security. A groundbreaking lattice-based digital signature technique is described in IACR TCHES by Ducas et al. The authors combine lattice-based encryption and other advanced cryptographic design methods to create a quantum-resistant signature scheme. The authors significantly advance post-quantum cryptography using trapdoor functions and lattice-based structures. This cryptographic approach is groundbreaking in the face of quantum computing threats. An extensive study on lattice-based digital signatures by Lyubashevsky et al.[8] included many advanced methods. Lattice reduction techniques, probabilistic polynomial time algorithms, security proofs made on hardness of lattice problems, among them SIS and LWE problems, and the asymptotic analyses of signature schemes are some of the methodologies used in the paper. Combining these methodologies, the authors had been able to give a wide-ranging look at lattice-based cryptography and shed light on its theoretical underpinning and practical efficiencies. J. Bos et al.[9] introduces "Frodo," a new quantum-secure key-exchange protocol based on LWE. It meticulously explores different models on key exchange mechanisms, parameter selection, security analysis, and practical implementations. Frodo innovatively applies LWE techniques for the achievement of robust quantum-resistant cryptography. It presents a promising avenue for secure communication in the quantum era. Erdem Alkim et al.[10] reviewed post-quantum key exchange protocols. The paper went into great detail in terms of models, such as Ring Learning With Errors, Ring-LWE, Ideal Lattices, and Kyber, to name but a few. Each protocol's merits and weaknesses were outlined in the paper. Post-quantum cryptography is explained in detail. M. R. Albrecht et al.[11] comprehensively review LWE and NTRU cryptographic algorithms. The report thoroughly examined those schemes' security, efficiency, and applicability. A detailed literature review, mathematical analysis, and actual experimentation are used. Cryptographic research benefits from their insights regarding scheme strengths and weaknesses. Albrecht et al.[12] performed a complete LWE concrete hardness investigation. The research tests many models to determine LWE solution difficulty. Many enumeration methods and lattice reduction algorithms like LLL, BKZ, and BKW are prominent models. Detailed examination of LWE-based cryptographic systems' difficulty provides invaluable insights for cryptographic protocol design and analysis.

R. Primas et al.[13] discuss "Single-Trace Side-Channel Attacks on Masked Lattice-Based Encryption". The paper proposes a method for analyzing side-channel vulnerabilities in masked lattice-based encryption schemes. The paper uses different methods, such as differential power analysis, correlation power analysis, and template attacks, to evaluate the security of these schemes. It reveals some weaknesses in masked lattice-based encryption against side-channel attacks. Tobias et al.[14] gives a comprehensive study on practical CCA2-secure and masked ring-LWE implementation. The authors delve into various cryptographic methods and techniques, such as ring-LWE, which is an important aspect of lattice-based cryptography. They study practical implementations of such techniques with a great emphasis on CCA2 security. The paper stands out for its detailed analysis of implementation strategies, providing insights into the achievement of robust security in embedded systems. Longa et al.[15] introduced a novel treatment for the acceleration of number theoretic transforms, or NTTs, for better ideal lattice-based cryptography. The Double-CRT technology, improved arithmetic, and careful memory management were among their many optimizations. By using these, they presented how one can achieve significant NTT computation speedups—essential for cryptographic schemes based on lattices—by using methodologies like the Cooley-Tukey algorithm and Montgomery reduction. All of these improvements combined point to significant increases in the performance of cryptographic schemes based on lattices.

### 3. Proposed Methodology

#### 3.1. NewHope

NewHope uses Ring-Learning-with-Errors (Ring-LWE) to exchange keys. It was submitted to the NIST Post-Quantum Cryptography (PQC) project and advanced to Round 2, but not Round 3. According to NIST's evaluation in NISTIR 8309, many of the core design elements of NewHope were absorbed into Kyber, which provided similar levels of security while offering more compact keys, smaller ciphertext sizes, and improved computational efficiency. In addition, deployment-oriented considerations such as reduced memory footprint and broader support within the cryptographic community favored Kyber as the stronger candidate for standardization. Thus, while NewHope was an important prototype that influenced the development of lattice-based KEMs, its ideas were ultimately carried forward into Kyber, which advanced further in the NIST PQC process. NewHope nonetheless remains a valuable reference design for understanding early implementations of Ring-LWE-based protocols [22].

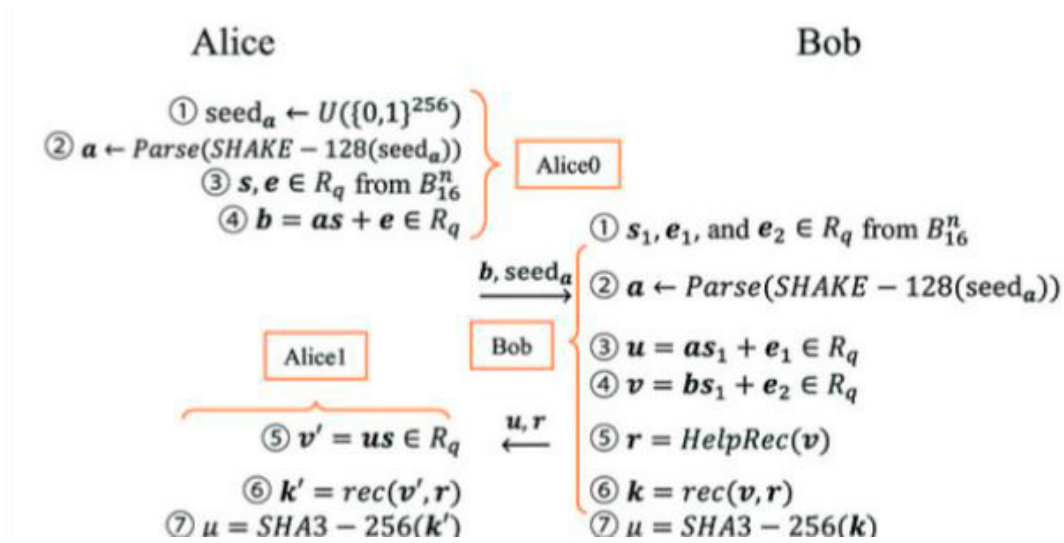


Fig. 1. NewHope

#### 3.2. Frodo

Conservative key-encapsulation systems like the FrodoKEM are practical post-quantum constructions. The security of these techniques depends on careful parameterization of the well-known learning with errors problem, which is closely related to conjectured-hard problems on generic, algebraically unstructured lattices. There are two variants of FrodoKEM: standard and brief. In standard use, there is no restriction on key pair reuse. In eFrodoKEM, a brief variant of FrodoKEM, a new key pair is created per protocol instance. German Federal Office for Information Security, BSI, recommends at levels 3 and 5 FrodoKEM as one of only two algorithms considered post-quantum and cryptographically suitable for long-term confidentiality [23].

#### 3.3. Kyber

NIST post-quantum cryptography standardization includes the lattice-based Kyber algorithm for encryption and key encapsulation. It is a theoretical construction of a cryptosystem that creates a public key from a polynomial matrix and a secret key from another polynomial matrix. To perform the encryption, the message of plaintext is encoded and

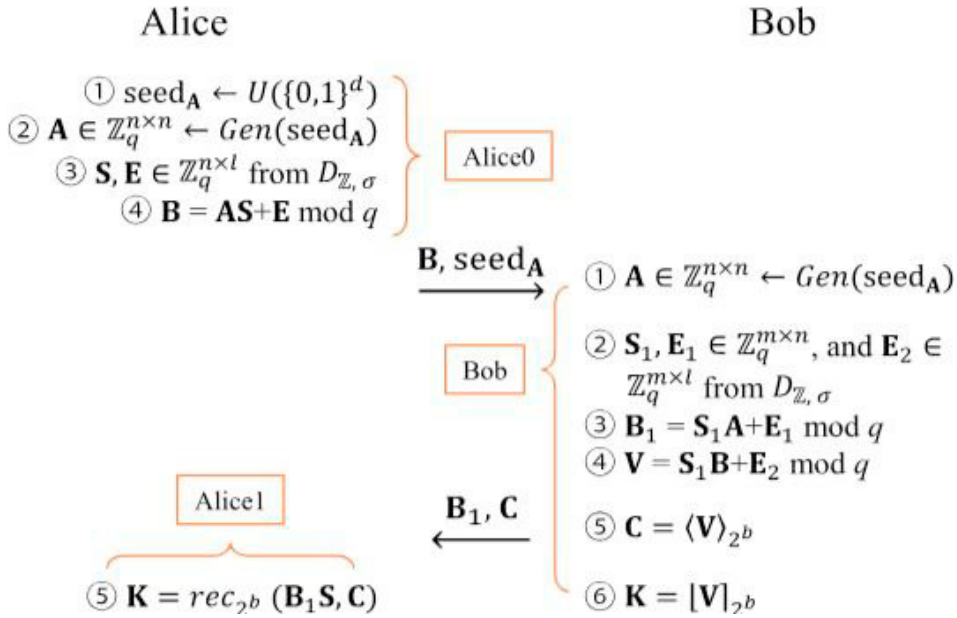


Fig. 2. Frodo

combined with the public key matrix and a random error polynomial to create the ciphertext. This utilizes a structure of module lattices to gain computational efficiency and security. The process of encryption ensures that the ciphertext is essentially indistinguishable from random noise—a vital property for secure communication.

Kyber decryption uses a private key matrix to remedy encryption errors. This stage retrieves the encoded message efficiently using polynomial arithmetic. After rectifying the error, reverse encoding yields the decoded message. The decryption process is efficient and secure, preserving data integrity and confidentiality [24].

### 3.4. Lizard

Lizard is an efficient, lightweight cryptographic technique for resource-constrained environments. It combines computational efficiency and robust security based on the Learning With Errors problem's hardness. Carefully specified parameters, modular arithmetic, and a discrete Gaussian error distribution characterize the approach. Implementation incorporates key generation—public and private keys, encryption—message encoding and ciphertext production, and decryption—error correction and message decoding. These processes ensure high security, maximum performance—computational efficiency and resource utilization—are the evaluation criteria. Performance also incorporates scalability—parameter variability and implementation flexibility. Lizard is secure against known attacks, computationally efficient, and applicable to diverse contexts. Thus, Lizard is a major advancement in lattice cryptography, providing a secure and efficient post-quantum solution. [25].

### 3.5. Ring-Lizard

To improve efficiency and security, Ring-Lizard adds ring-based structures, specifically the Ring Learning With Errors (Ring-LWE) problem, to the Lizard algorithm. It uses ring arithmetic to compute quicker while maintaining lattice-based cryptography's security in resource-constrained contexts.

The Ring-Lizard encryption procedure begins with key generation, which generates the public key from a ring-based polynomial, making the secret key computation too difficult. A secret polynomial makes up the private key. Encryption converts plaintext into a polynomial representation for ring operations. The ciphertext is created by adding the encoded message to the public key and random error polynomials. This phase greatly improves computing performance by using ring structures to speed up and apply encryption.

Ring-Lizard decrypts by correcting encryption errors with the private key polynomial. This phase uses the ring's

algebraic property to efficiently recover the encoded message. After fixing the error, the polynomial yields the decoded message. Decryption ensures data integrity and confidentiality by accurately and securely recovering the message. [26].

#### 4. Results and Discussion

Key generation, encryption, and decryption times are used to compare lattice-based encryption methods. Key generation time is the time needed to generate cryptographic keys for communication and data security. This depends on method complexity, key size, and computer power. The encryption time for a given algorithm and input size defines the duration it would take for that algorithm to convert plaintext data into a cryptogram that is secure for either communication or storage. The variance in decryption time is based on factors such as the complexity of the encryption algorithm, length of the key, and computational power, but generally, the figure ranges from milliseconds to years depending on the specifics of the encryption. Based on the result from Table I we can observe that kyber outperforms another lattice-based encryption algorithm.

| Algorithms  | Key Gen Time | Encryption Time | Decryption Time |
|-------------|--------------|-----------------|-----------------|
| Kyber       | 5.86ms       | 4.77ms          | 0.79ms          |
| Ring Lizard | 228.30ms     | 41.26ms         | 1.78ms          |
| Lizard      | 986.67ms     | 7.03ms          | 2.33ms          |

Table 1. Comparison time of the three lattice algorithms

The comparison of the lattice based key exchange protocols are based on two evaluation metrics which are Total Key generation time and Key exchange time. Total key generation is the amount of time used in generating cryptographic keys to ensure that communication or the encryption of data is done safely, so it can help in creating and maintaining an assurance of confidentiality, integrity, and authenticity in digital transactions and communications.

| Key Exchange | Total Key generation time | Key exchange time |
|--------------|---------------------------|-------------------|
| Frodo        | 171.51 ms                 | 0.0011 ms         |
| New hope     | 81.924 ms                 | 0.084 ms          |

Table 2. Comparison time of the two key exchange protocols

Key exchange time is the time taken for cryptographic protocols to establish shared encryption keys between parties securely; this goes toward ensuring confidentiality and integrity in communication. Based on these two evaluation metrics we come to observe that NewHope key exchange protocol takes significantly less time to generate key and takes lesser time to exchange the key. This can be observed from Table II.

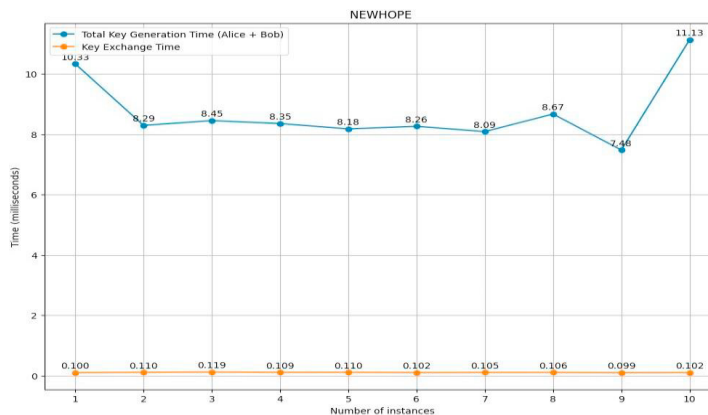


Fig. 3. NewHope Line Graph

#### 4.1. NewHope

The line graph in Fig 3 shows the total key generation time of the NewHope key exchange protocol for different numbers of instances. The x-axis represents the number of instances of a key generator, ranging from 1 to 10. The y-axis shows time in milliseconds, with a range of 0 to around 70 milliseconds. There are two lines on the graph, each representing a different measurement: Key Exchange Time and Total Key Generation Time (Alice + Bob).

The data points on the line represent the total key generation time for a certain number of instances. For example, for 2 instances, it took approximately 8.3 milliseconds to generate keys, and for 10 instances, it took approximately 11.1 milliseconds. Overall, the graph shows that the total key generation time increases as the number of instances increases. This suggests that it takes longer to generate keys when more instances of the NewHope protocol are involved.

The increase in key generation time with more instances suggests the NewHope protocol might not be highly scalable. While the increase seems small (around 3 milliseconds from 2 to 10 instances), it could become significant for a much larger number of instances. The relatively low total key generation time (under 12 milliseconds for 10 instances) suggests the protocol might be efficient for smaller applications or those not requiring super-fast key exchange.

#### 4.2. Frodo

Fig 4 shows the line graph for the key exchange protocol named Frodo. The x-axis represents the number of instances of a key generator, where it ranges between 1 and 10. The y-axis represents time in terms of milliseconds,

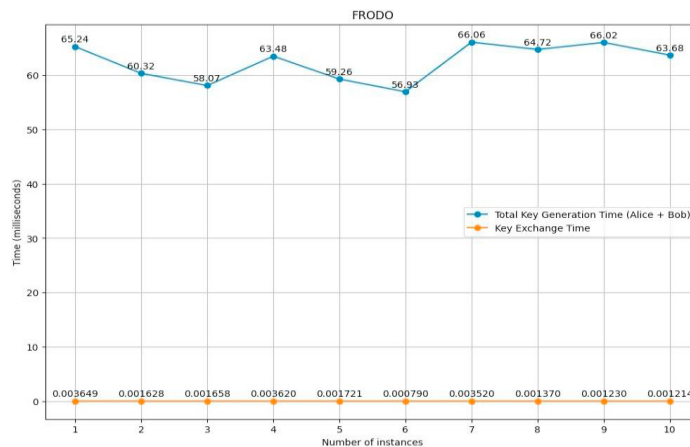


Fig. 4. Frodo Line Graph

where it ranges between about 0 and up to around 70 milliseconds. On this graph, there are two lines. One measurement for key exchange time is represented; another is total key generation time for Alice plus Bob.

The total key generation time for both parties increases with an increase in instances of a key generator. This indicates that it takes longer to generate keys if there are many instances. The key exchange time fluctuates more than the total key generation time.

#### 4.3. Kyber

The line graph in Fig 5 shows the performance of the Kyber lattice algorithm for different numbers of instances. The graph's x-axis indicates incidences from 1 to 10. The y-axis shows milliseconds for each measurement type.

**Key Generation Time:** This line represents the Kyber lattice method key pair generation time. The time increases from 0.8 milliseconds for 1 instance to 24 for 10 instances.

**Time to encrypt a message using the Kyber lattice technique.** As the number of instances increases, encryption time increases from 2.8 milliseconds for 1 instance to 12 for 10.

**Time to decrypt a message using the Kyber lattice method.** Decryption time increases slightly with instance count,

but less than key generation and encryption time. It ranges from 3.5 to 4.5 milliseconds for 10 occurrences.

Overall, the graph shows that the Kyber lattice algorithm takes longer to perform all three cryptographic tasks (key generation, encryption, decryption) as the number of instances increases. This suggests that the algorithm's performance scales with the number of instances. The decryption time seems to be less affected by the number of instances compared to key generation and encryption. This suggests a potential trade-off between security (potentially enhanced with more instances) and decryption efficiency.

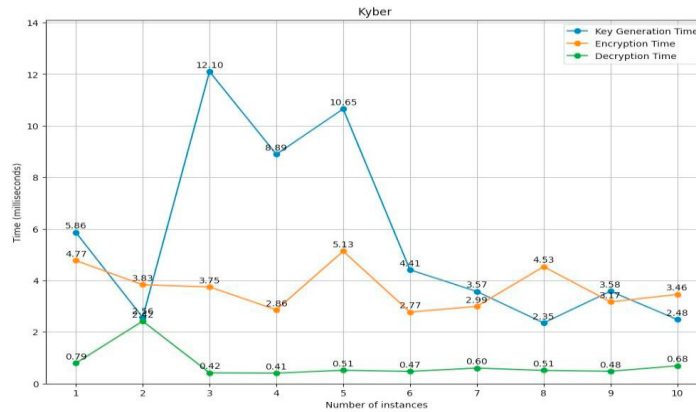


Fig. 5. Kyber Line Graph

#### 4.4. Lizard

Fig 6 depicts the line graph that shows the time it takes to complete different cryptographic tasks for a certain number of lizard instances. The x-axis of the graph shows the number of ring lizard instances, which goes from 2 to 10. The y-axis represents milliseconds.

Three lines indicate cryptographic tasks in the graph:

- Key Generation Time
- Encryption Time
- Decryption Time

The graph illustrates the cryptographic task completion time for each ring lizard instance. Two lizard instances required 6 milliseconds to produce a key, 10 milliseconds to encrypt, and 6 milliseconds to decrypt. Overall, the graph illustrates that cryptographic tasks take longer as lizard instances rise. This implies that the cryptographic system is increasingly resource-intensive with more lizard instances. With additional lizard instances, key generation, encryption, and decryption take longer, suggesting the approach may not scale effectively. As cases rise, key creation also increases in time, which is concerning. It's hard to tell from the graph, but key generation time seems to be most affected by instances. This may indicate that Lizard lattice key generation is computationally expensive and scales poorly with many instances.

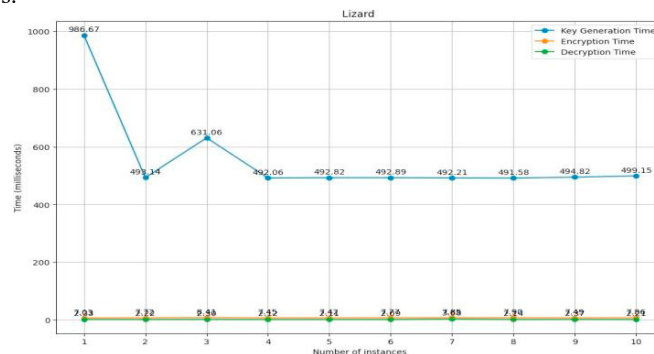


Fig. 6. Lizard Line Graph



#### 4.5. Ring Lizard

Fig 7 shows ring lizard instances' cryptographic task completion times as a line graph. The graph's x-axis shows 1-10 ring lizards. The y-axis shows milliseconds. Three lines in the graph show key generation, encryption, and decryption times. The graph illustrates the cryptographic task completion time for each ring lizard instance. When using 2-ring lizard instances, it took 6 milliseconds to generate a key, 10 to encrypt, and 6 to decrypt. Each cryptographic task takes longer as ring lizard instances multiply. To construct a key for 2 ring lizard instances took 6 milliseconds, but for 10 it took 24. Overall, the graph illustrates that cryptographic tasks take longer when ring lizard instances rise. This implies that the cryptographic system is more resource-intensive with more ring lizard instances.

#### 4.6. Performance Measurement Methodology

All schemes were benchmarked in a consistent hardware and software environment to assure performance reliability. The trials were ran on a PC with an Intel Core i7 processor, 16 GB RAM, Ubuntu 22.04 LTS, and Node.js v18. Each measurement was repeated 50 independent runs, and the reported values represent the average execution time to minimize the effect of outliers caused by runtime fluctuations. For each cryptosystem, standard parameter sets recommended in the respective specifications were used (e.g., polynomial degrees, modulus sizes, and error distributions). For NewHope and Frodo, we adopted the parameterizations aligned with their NIST submissions, while Kyber, Lizard, and Ring-Lizard were evaluated using security parameters targeting the equivalent of NIST Level 1–3 security.

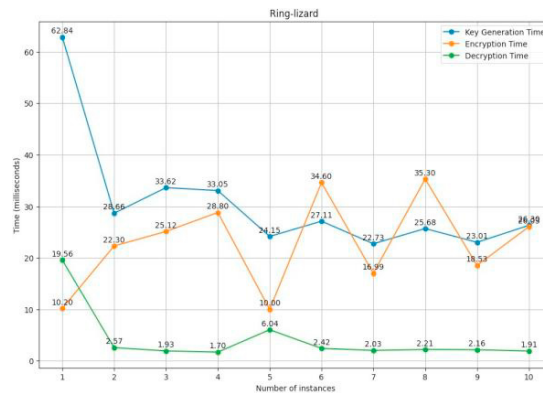


Fig. 7. Ring Lizard Line Graph

The evaluation metrics include key generation time, encryption time, and decryption time for encryption schemes, and total key generation time and key exchange time for key exchange protocols. Results were normalized across all runs to provide consistent comparisons. This methodology ensures that the performance results presented in Tables I and II accurately reflect practical execution costs and can be directly compared with published benchmarks.

#### 4.7. Security Trade-offs and Deployment Considerations

Beyond raw performance, the choice of a lattice-based cryptosystem also involves significant trade-offs between security, efficiency, and practical deployment. Kyber demonstrates the most favorable balance, offering compact key sizes and fast execution, which explains its advancement in the NIST PQC process. In contrast, Frodo is regarded as a more conservative design since it is based on generic, unstructured lattices; while this enhances confidence in its long-term security, it comes at the cost of larger key sizes and slower execution times. NewHope, though efficient, was ultimately not selected for later NIST rounds because its design contributions were largely absorbed into Kyber, which provided similar security with improved efficiency and memory use (NISTIR 8309). From a security perspective, lattice-based schemes are conjectured to be resistant to quantum attacks; however, they remain vulnerable to implementation-level threats. Studies such as Primas et al. [13] demonstrate that single-trace side-channel attacks can compromise masked lattice-based schemes, underscoring the importance of constant-time coding practices, careful parameterization, and side-channel countermeasures.

Deployment considerations are equally critical. Lightweight algorithms such as Lizard and Ring-Lizard may be particularly relevant for IoT and constrained environments, where computational and memory resources are limited. On the other hand, JavaScript-based implementations, while useful for demonstrating cross-platform applicability, are inherently constrained by client-side execution time and memory availability in browsers. Finally, real-world adoption will depend on interoperability with standardized algorithms, particularly those chosen by NIST, such as Kyber for key encapsulation and Dilithium for digital signatures, to ensure long-term security and ecosystem compatibility.

#### 4.8. Comparative Analysis with Literature

To validate our findings, we compared our benchmark results with those reported in related works. Our measured performance of Kyber (5.86 ms key generation, 4.77 ms encryption, 0.79 ms decryption) closely aligns with the benchmarks reported in Bos et al. [9] and Alkim et al. [10], where Kyber consistently demonstrated superior efficiency among lattice-based schemes. Similarly, the trends observed for Frodo and NewHope are consistent with NIST's Post-Quantum Cryptography evaluations (NISTIR 8309, 2020). Frodo was confirmed to be conservative and highly secure, though at the expense of larger keys and slower performance, while NewHope offered faster execution but was eventually integrated into Kyber, which advanced further in the standardization track.

### 5. Conclusion

Proposed implementation and benchmarking demonstrate that lattice-based schemes are practical for post-quantum secure communication, particularly on JavaScript platforms relevant to web and IoT applications. Kyber achieved the best balance between speed and efficiency, while Frodo exemplified a conservative design suitable for long-term confidentiality. NewHope offered valuable insights despite its exclusion from later NIST rounds. Lizard and Ring-Lizard showed promise for lightweight, resource-constrained environments. This study highlights not only performance metrics but also the broader trade-offs among security, efficiency, and deployment feasibility. Future work will focus on integrating side-channel countermeasures, extending comparisons to other NIST finalists such as SABER, and evaluating cross-platform performance.

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